Number Codes (Set B)

Year 4 · **Homework Worksheet**

Worked Example

Example: A code doubles a number and then adds 2. What is the code for 9?

Answer: 20 — 9 × 2 = 18, then 18 + 2 = 20.

Purple — Accessible (15 questions)

1. A diary lock uses this code: 1 = B, 4 = E, 9 = S. What is the code for the number 149?
2. A board game spinner uses this code: 2 = R, 3 = U, 5 = N. The code RUN stands for which number?
3. A bike chain lock uses this code: 1 = S, 4 = U, 6 = N. What is the code for 146? A) NUS B) USN C) SUN D) UNS E) NSU
4. An arcade token machine uses this code: 0 = F, 3 = A, 7 = R. What is the code for the number 730?
5. A museum keypad uses this code: 1 = D, 4 = O, 8 = G. The code OGD stands for which number?
6. A shed padlock uses this code: 2 = C, 3 = A, 7 = R. The code CAR stands for which number? A) 237 B) 273 C) 327 D) 372 E) 723
7. A campsite gate uses this code: 2 = T, 5 = E, 6 = N. What is the code for the number 265?
8. A science lab door uses this code: 1 = W, 3 = I, 7 = N. The code WIN stands for which number?
9. Using the code 0 = J, 1 = T, 2 = O, 3 = P, 4 = N, 5 = W, 6 = F, 7 = S, 8 = L, 9 = M, select ALL the numbers below whose code would END in the letter 'T'. A) 341 B) 472 C) 918 D) 251 E) 501
10. A tent zip uses this code: 1 = T, 4 = O, 9 = P. What is the code for 491? A) TOP B) POT C) TPO D) OTP E) OPT
11. A football scoreboard uses this code: 0 = G, 4 = O, 8 = L. What is the code for the number 408?
12. A train platform display uses this code: 2 = F, 3 = A, 5 = R. The code AFR stands for which number?
13. A diary padlock uses this code: 2 = O, 5 = L, 6 = D. The code OLD stands for which number? A) 265 B) 256 C) 526 D) 625 E) 652
14. A library card uses this code: 1 = C, 5 = A, 6 = T. What is the code for the number 516?
15. A space rocket control panel uses this code: 2 = M, 6 = O, 7 = N. The code MON stands for which number?

Green — Standard (15 questions)

1. A code is formed by doubling a number and then adding 4. What is the code for 5?
2. A code is formed by multiplying a number by 3 and then adding 1. The code for a number is 25. What was the original number?
3. A code is formed by doubling a number and then adding 7. What is the code for 6? A) 17 B) 18 C) 20 D) 19 E) 16
4. A code is formed by adding 2 to a number and then multiplying the result by 3. What is the code for 4?
5. A code is formed by halving a number and then adding 3. The code for a number is 11. What was the original number?
6. A code is formed by multiplying a number by 4 and then subtracting 3. The code for a number is 25. What was the original number? A) 6 B) 8 C) 7 D) 9 E) 5

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7. In a code, 2 → 9 and 4 → 17. Using the same rule, what is the code for 6?

8. In a code, 3 → 10 and 6 → 19. If the code for a number is 16, what is that number, using the same rule?

9. A code is formed by doubling a number and then adding 3. Select ALL the numbers below whose code would be LESS than 20. A) 6 B) 7 C) 8 D) 9 E) 10

10. In a code, 3 → 13 and 6 → 25. Using the same rule, what is the code for 8? A) 33 B) 32 C) 34 D) 31 E) 35

11. A code is formed by subtracting 1 from a number and then multiplying by 5. What is the code for 5?

12. A code is formed by multiplying a number by 2 and then subtracting 5. The code for a number is 9. What was the original number?

13. A code is formed by multiplying a number by 3 and then subtracting 4. What is the code for 7? A) 16 B) 17 C) 18 D) 19 E) 15

14. In a code, 4 → 14 and 7 → 23. Using the same rule, what is the code for 9?

15. In a code, 2 → 7 and 5 → 16. If the code for a number is 22, what is that number, using the same rule?

Standard / Advanced (15 questions)

1. In a code, every digit has 3 added to it, wrapping after 9. What is the code for 481, and is that code an even or odd number?

2. In a code, every digit has 6 added to it, wrapping after 9. What is the code for 273, and what is the sum of its three digits?

3. A code reverses the digits of a number. For 583, how many digits stay in exactly the same position after reversing? A) 0 B) 1 C) 2 D) 3 E) 4

4. In a code, every digit has 4 added to it, wrapping after 9. The code for a number is 206. Is the original number bigger or smaller than 206?

5. A code reverses the digits of a number. If the code for a number is 918, what is the SUM of the original number's digits?

6. In a code, every digit has 5 added to it, wrapping after 9. What is the code for 347?

7. Using the same rule (add 5, wrap after 9), the code for a number is 492. What was the original number?

8. A code reverses the digits of a number. What is the code for 731? A) 137 B) 173 C) 317 D) 713 E) 371

9. A number's digits are flipped end to end to make its code. The code 624 came from which starting number?

10. In a code, every digit has 8 added to it, wrapping after 9. Select ALL the numbers below whose code would contain a 0. A) 251 B) 162 C) 374 D) 481 E) 593

11. In a code, every digit has 9 added to it, wrapping after 9. What is the code for 456?

12. Using the same rule (add 9, wrap after 9), the code for a number is 678. What was the original number?

13. A code reverses the digits of a number. What is the code for 905? A) 509 B) 590 C) 095 D) 950 E) 059

14. In a code, every digit has 1 added to it, wrapping after 9. What is the code for 892?

15. Using the same rule (add 1, wrap after 9), the code for a number is 710. What was the original number?

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